

Element A: Presentation and Justification of the Problem

A middle school librarian wants to create a “This or That” Challenge station near the checkout line where students can vote between two fun choices, such as “Cats or Dogs?” or “Pizza or Tacos?” The purpose of the station is to entertain students while they wait, encourage interaction, and create easy conversation starters between students and staff. The station must be compact enough to fit on the librarian’s counter, but also large and visible enough for many students to participate by placing items such as pompoms, marbles, or tokens into containers representing their choices. The design also needs to be durable because it will be handled daily by many middle school students. The materials must also be harmless if broken (i.e. glass is not a suitable solution as glass shards could potentially harm the children). This challenge addresses asocial issue within schools: helping students feel more comfortable communicating with others and teachers. Middle school can be a difficult social environment, especially for shy or introverted students. According to the National Institute of Mental Health, many adolescents experience social anxiety, which can make everyday interactions stressful or uncomfortable (National Institute of Mental Health, “Social Anxiety Disorder”). Many introverted or shy kids might find it hard to start a conversation with people. Some kids might want to talk to people or teachers but do not know what to say. They might even avoid people or social conversations, fearing that they’ll be judged. A station like this could lower the pressure and help the librarian herself to find something to talk to students, even if it's a small topic like the students' choice.

The librarian is one of the primary stakeholders because she wants students to feel joy while visiting the library. Libraries are not only for reading or studying, but also community spaces where students can feel safe. Research from the American Library Association explains that school libraries help support students’ well-being and foster a nurturing environment (American Library Association, “Literacy and wellbeing at school libraries”). A conversation-based activity station could help the librarian interact more easily with students, which could encourage students to spend more time in the library environment.

The middle school students are another stakeholder group as they are the ones using the device. Many students experience stress, loneliness, or difficulty making friends. The Youth Risk Behavior Survey Data Summary and Trends Report shows that around 42% students feel a sense of hopelessness almost every day (CDC, “Youth Risk Behavior Survey Data Summary & Trends Report”). Positive in-person interactions are important for a student’s mental health, especially during this rise of new technology.

The school staff are also stakeholders because student engagement can contribute to a more positive school atmosphere. If students feel more comfortable interacting with other students and teachers, they may also become more comfortable asking questions, participating in class, and seeking help when needed.

Parents are indirect stakeholders because they want schools to support both the academic and emotional growth of their children. In the article, “How to Cultivate a Sense of Belonging in

Schools,” the author writes, “belonging also impacts mental wellness (a lack of belonging can cause depression and antisocial behavior), affects academic performance (course persistence and grades), and even has lifetime benefits (such as career satisfaction and community engagement)” (Edutopia, “How to Cultivate a Sense of Belonging in Schools”). Additionally, the school itself benefits from creating welcoming spaces.

This problem is important because student mental health and social connection have become growing concerns in schools. Studies show that adolescents are experiencing increasing rates of loneliness, anxiety, and isolation, especially after the COVID-19 pandemic (National Library of Medicine, “Loneliness before and after COVID-19: Sense of Coherence and Hope as Coping Mechanisms”). By creating small, low-pressure social interactions, this could help students feel more comfortable and connected in school. While it may seem minute, a simple “This or That” Challenge could possibly encourage hundreds of students to interact with each other and teachers. By promoting conversation and interaction in a fun and accessible way, this project could help improve the overall school experience for many students.

Miroboard: miro.com/app/board/uXjVGpkaPUU=

Problem Definition:

A library needs a "This or That" challenge that students can participate in while they wait in line.

Prototype sketch:

First design CAD:

First design prototypes:

Final iteration prototypes:

Final iteration CAD:

Iteration Changes:

We decided to add Velcro strips on the base and the bottom of the cups. We realized that it's very difficult to empty the cups without having to turn the entire device over so our iteration will allow the cups to be detached and reattached from the base. The second iteration we decided to add question cards on the cups. We realized that it was pretty hard to see which option is each of the cups, so we decided to make it clear what cup is the first option and what cup is the second. These changes still keep our core simple design but add on it.

Testing Plan:

Same as last time

Test #	Test Description	Pass/Fail Criteria
1	Can the prototype hold two cups without falling over?	Pass: Prototype holds two cups without falling over. Fail: Prototype falls over or cups are not held.
2	Can the prototype hold question cards on the cups?	Pass: Question cards are held on the cups. Fail: Question cards are not held or are difficult to read.
3	Can the prototype be easily moved and reattached?	Pass: Prototype can be moved and reattached easily. Fail: Prototype is difficult to move or reattach.
4	Can the prototype be used by multiple people?	Pass: Multiple people can use the prototype. Fail: Prototype is only usable by one person.
5	Can the prototype be used in a library setting?	Pass: Prototype is suitable for library use. Fail: Prototype is not suitable for library use.

Feedback:

Feedback from the testing process was positive. The prototype was easy to use and the question cards were clear. The Velcro strips were a helpful addition. The design was simple and functional. The prototype was well-received by the students and the library staff. The design was easy to build and the materials were readily available. The prototype was a success and the design was a good fit for the library setting.

Presentation Video:

<https://docs.google.com/videos/d/14PLoEs9P-lazyfzRce2pkACFo-v8MO6pNbCmn8wBNU/edit?scene=id.p#scene=id.p>

Most of our tests passed. However, due to the change of materials, a few tests, such as test 4 and 5, did not work.

Sources:

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